

Autonomous Wall Follower

(the first lidar project: “lidar stop with remote”, and this wall following project and the final projects can be demo’d by Monday and Wednesday April 27th and 29th) All late projects after April 29th will be demo’d during our allotted final exam time (May 4th 12pm).

Core idea for this next project:

The robot should drive on its own while maintaining a target distance from a wall on either the left or right side.

This teaches:

- closed-loop control
- sensor interpretation
- reactive autonomy
- state-based behavior

What the robot should do

You should write a program so the robot can:

- move forward autonomously
- detect a wall on one chosen side
- stay a set distance from that wall
- avoid crashing into obstacles in front
- recover if the wall is temporarily lost

This should be a clean jump from the LIDAR safety-stop project.

Project structure

Basic version

The robot should:

- look at the front distance
- look at the wall-side distance

- try to keep the wall-side distance near a target value, put upper and lower thresholds so your robot is not having spasms.
- turn away if it gets too close to the wall
- turn towards if it gets too far from the wall
- stop or sharply turn if something is directly in front

Zones

You can divide the LIDAR into simple angle regions.

Example for **right-wall follow**:

- **front**: 340–20 degrees
- **right**: 250–290 degrees
- maybe **front-right**: 290–340 degrees, use front-right and back-right angles so you can track the wall better.

You do not need fancy mapping. Just use minimum or average distance in each zone.

Simple behavior logic

Case 1: obstacle in front

If front is too close:

- stop forward motion
- turn left until front clears

Case 2: wall is too close

If distance is smaller than target:

- steer away a little

Case 3: wall is too far

If distance is larger than target:

- steer towards a little

Case 4: no wall detected on right

If the right side suddenly becomes very open:

- gently turn right to search for the wall

That is the complete project.

Keep your journal logs and turn those in when you demo. All demos of this project must be done by Monday April 27th but remember there is a final project that will be due at the same time and it will .